

ECONOMICS OF EEELECTRIC COOKING IN NEPAL

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IAEE 2025 Paris Conference

15 - 18 June 2025

Paris, France

Introduction (1/2)

- ❖ Residential sector is largest energy consumer (61% of the total), and more than half households use traditional biomass for cooking;
- ❖ WHO estimates about 25 thousand deaths (2019 data) due to household air pollution, most of them were children and women;
- ❖ Nepal's electricity supply is entirely clean/renewables (hydro), yet only 0.5% of households use electricity for cooking;
- ❖ The country has surplus electricity in the wet season, and power producers are compelled to shut down due to a lack of demand;
- ❖ Imported LPG is subsidized; fossil fuels share highest in the import bills;
- ❖ Switching to electric cooking could benefit the country from economic, health, and environmental perspectives;

Introduction (2/2)

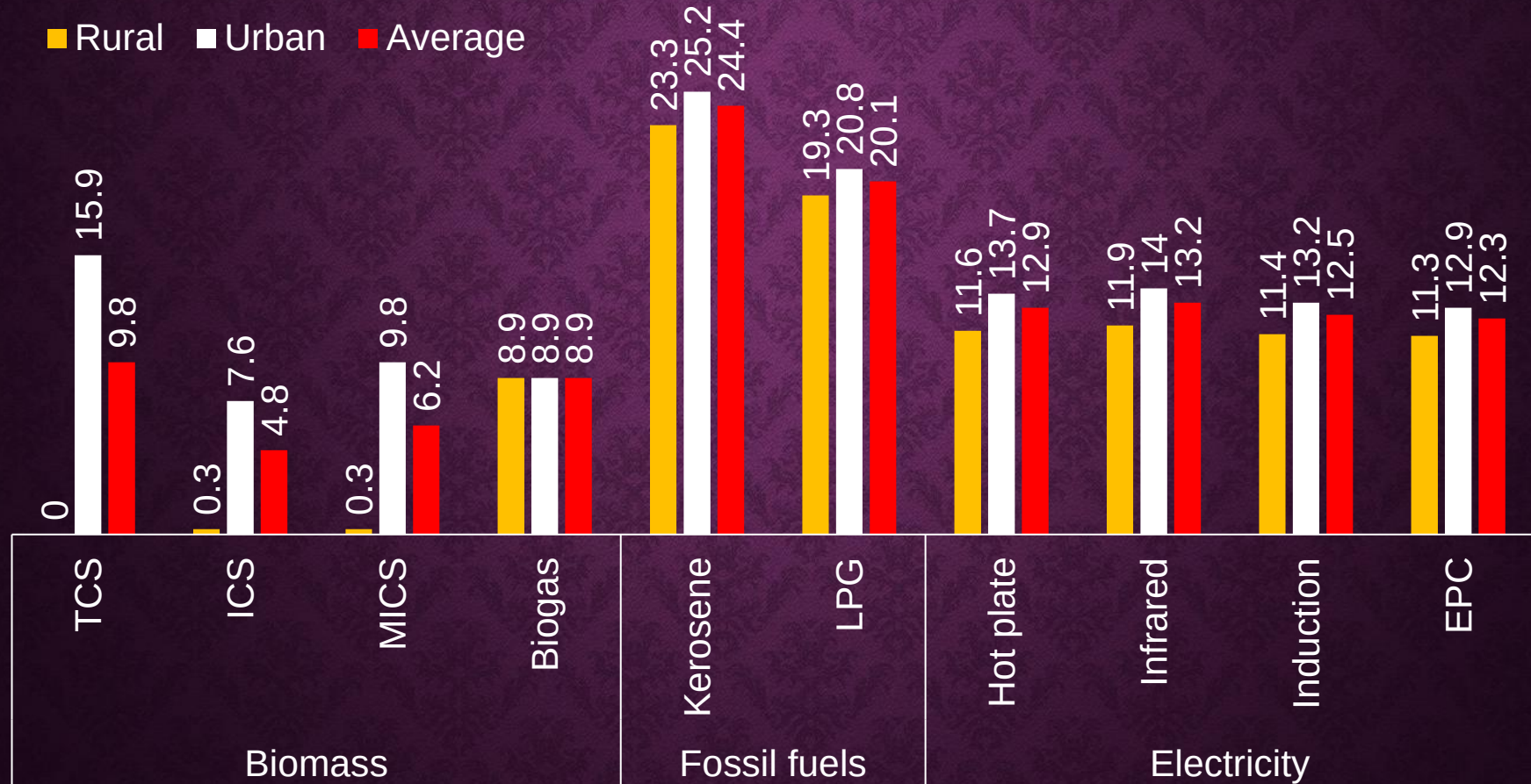
- ❖ First, we analyze the economics of various cooking alternatives for households from private and social perspectives at different geographical regions
- ❖ Second, it estimates the wider economic impacts of substituting imported LPG with domestically produced hydropower for household cooking.

Methodology and Data

- ❖ We developed an excel based economic model for calculating the levelized costs of cooking (LCOC) using various fuels/technologies in households;
- ❖ LCOC accounts for not only the cost of technologies and fuels but also average size of households/meal size and meal preparation frequency;
- ❖ The model calculates the costs from (i) private and (ii) social perspectives;
- ❖ We also estimate wider economic impacts of switching imported LPG with domestically produced clean electricity for HH cooking;
- ❖ Data were collected through primary and secondary sources.

Results: Private costs of cooking (1/4)

Average costs of cooking from private perspective, Country Level
(‘000 NRs/household/year)



TCS, ICS and MICS are respectively traditional, improved and metallic stoves;
EPC is electric pressure cooker

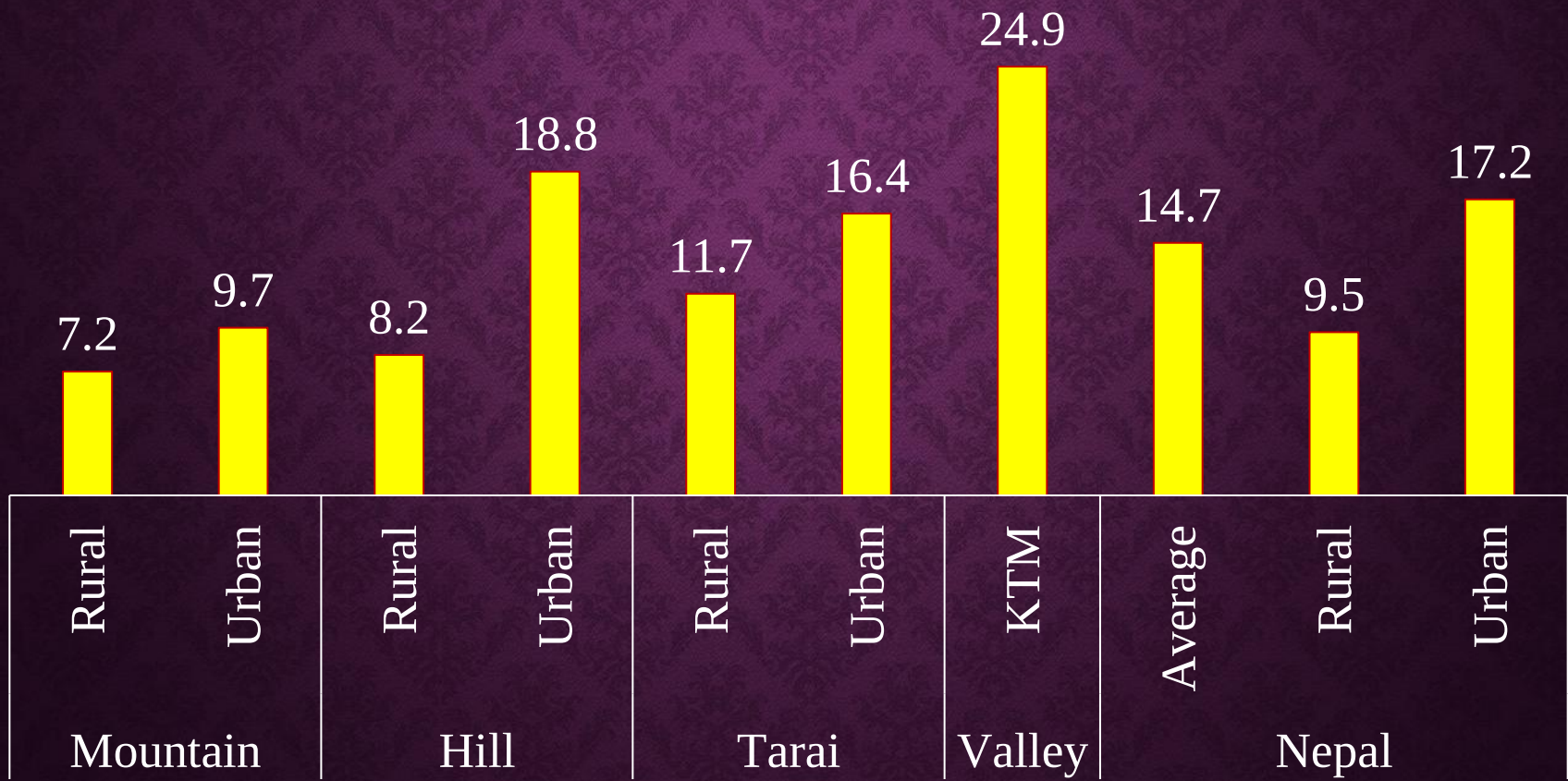
Results: Private costs of cooking (2/4)

Average costs of cooking from private perspective, Regional level
(‘000 NRs/household/year)

		Mountain		Hill		Tarai		Kathmandu Valley
		Rural	Urban	Rural	Urban	Rural	Urban	
Biomass	TCS	0	18.7	0	11.5	0	17.1	28.7
	ICS	0.3	8.9	0.3	5.6	0.3	8.1	13.4
	MICS	0.3	11.5	0.3	7.2	0.3	10.5	17.4
	Biogas	8.9	8.9	8.9	8.9	8.9	8.9	8.9
Fossil fuels	Kerosene	29.6	31.7	20.9	19.7	22.1	23.3	19.7
	LPG	24.2	25.9	17.4	16.5	18.3	19.3	16.4
Electricity	Hot plate	12.6	13.5	10.7	12.5	12.7	14.6	12.4
	Infrared	12.9	13.8	11.1	12.8	13.1	15	12.8
	Induction	12.3	13	10.6	12.2	12.4	14.1	12.1
	EPC	12.1	12.8	10.5	12	12.2	13.7	11.9

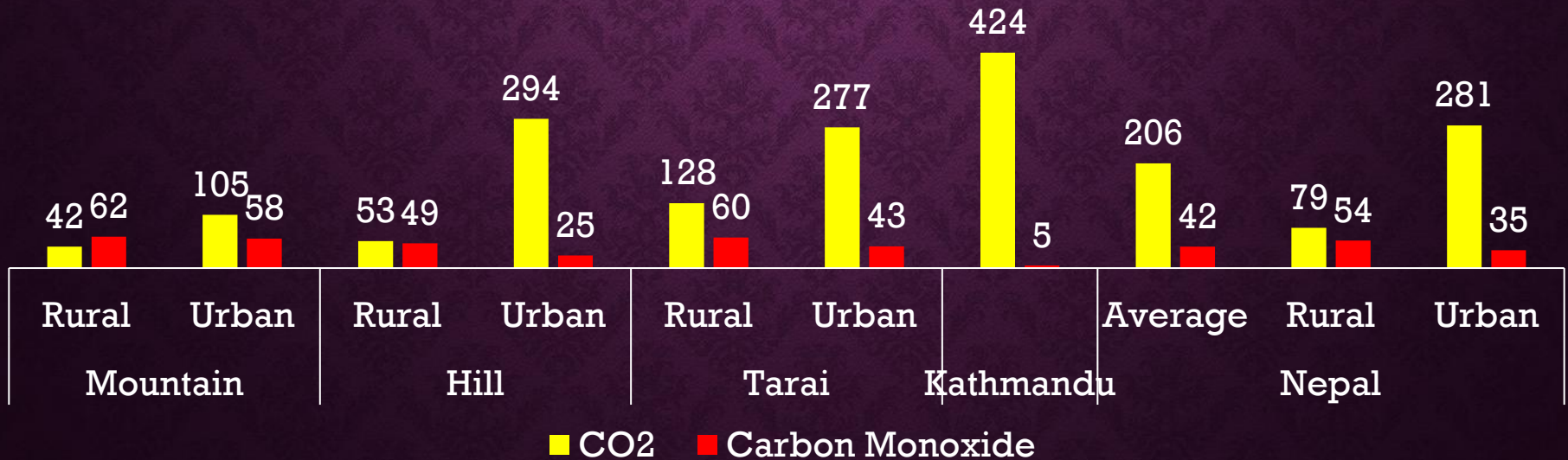
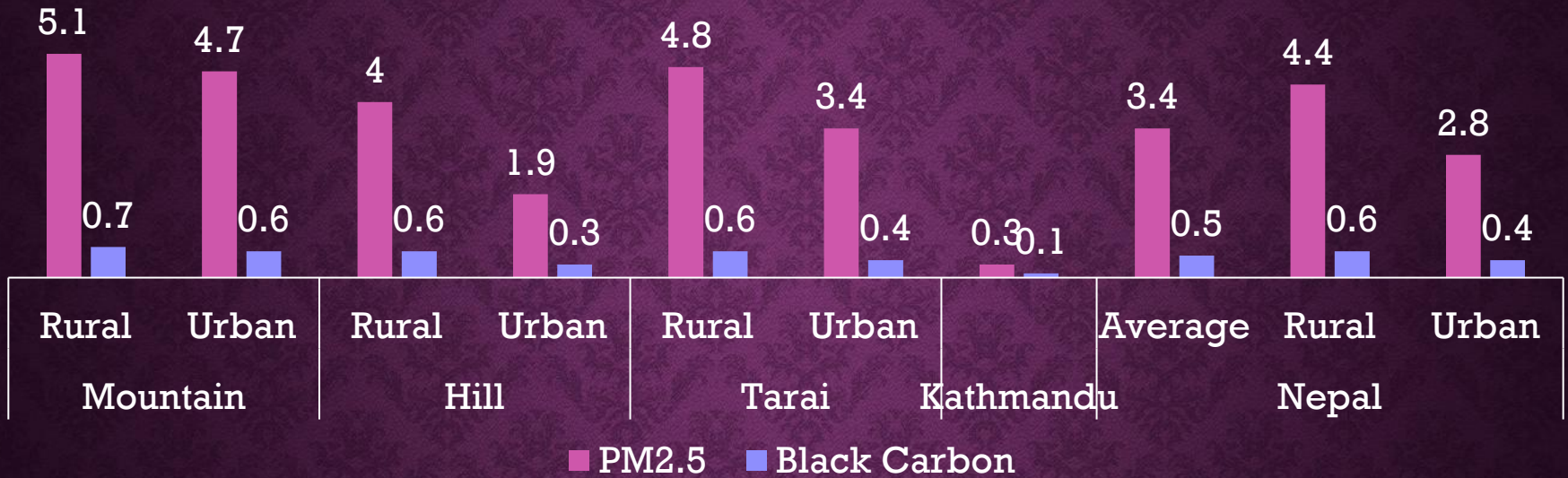
Results: Private costs of cooking (1/4)

Average costs of cooking from private perspective, Country Level
With Fuel Stacking ('000 NRs/household/year)



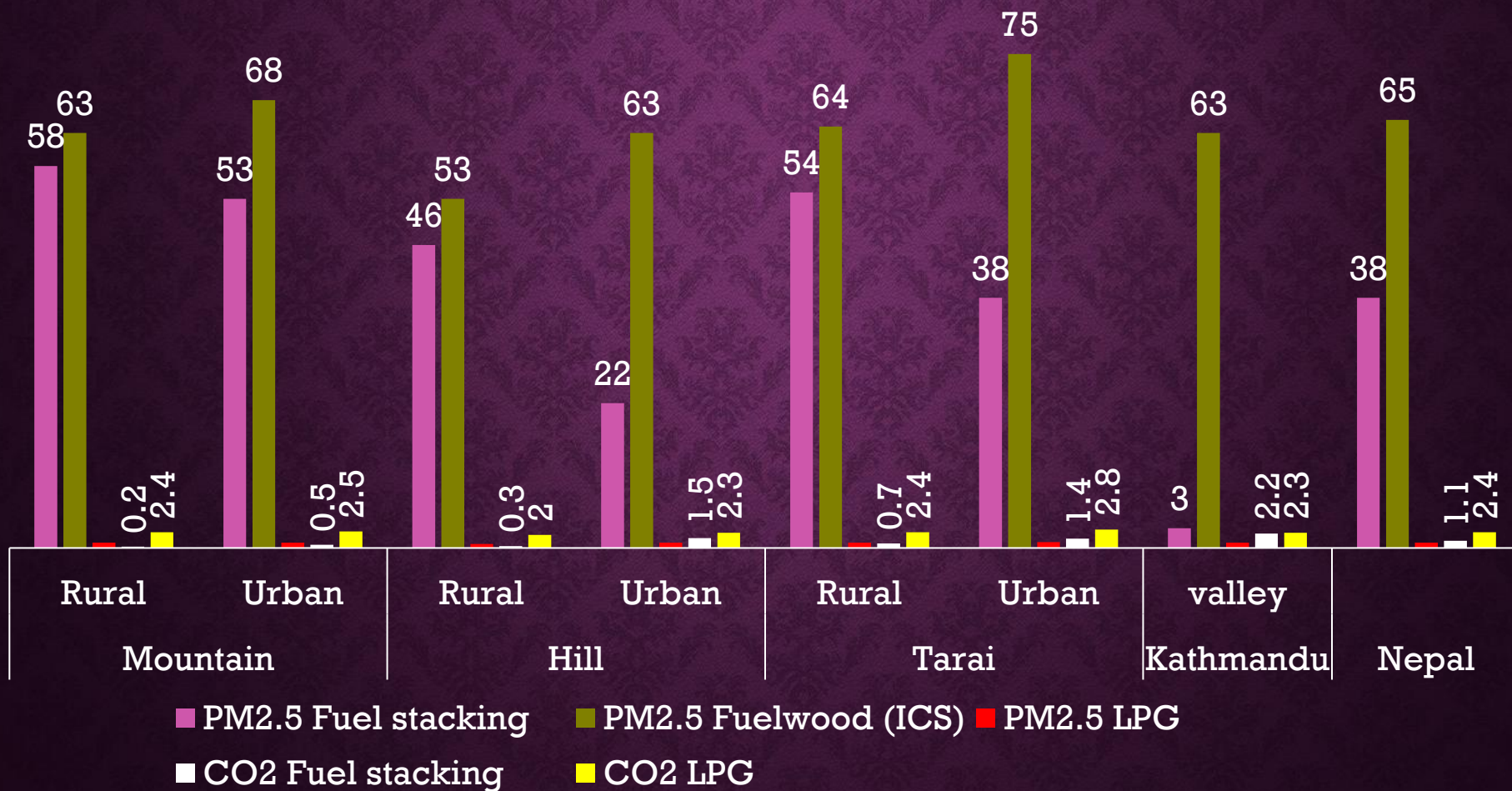
Results: Emissions

Annual emissions (kg/household/year)



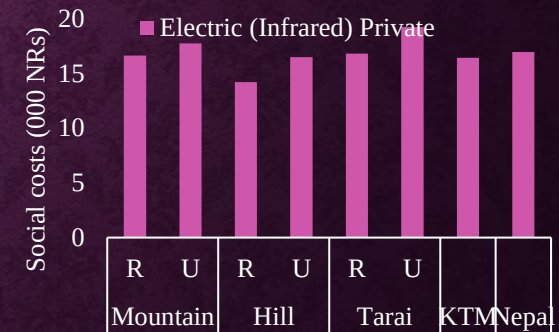
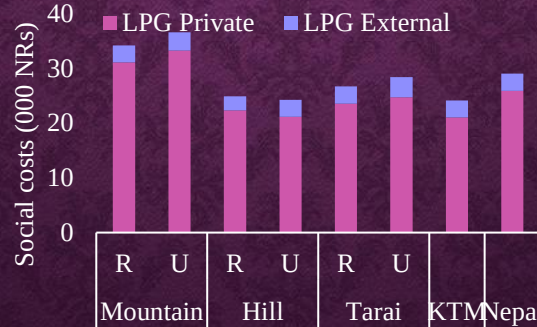
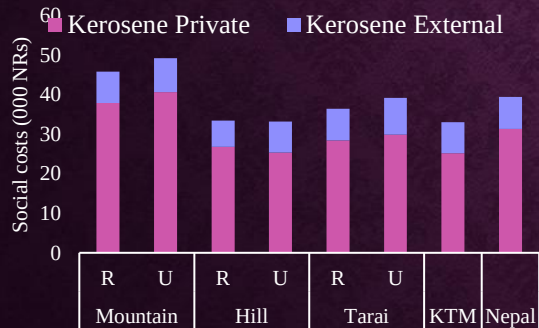
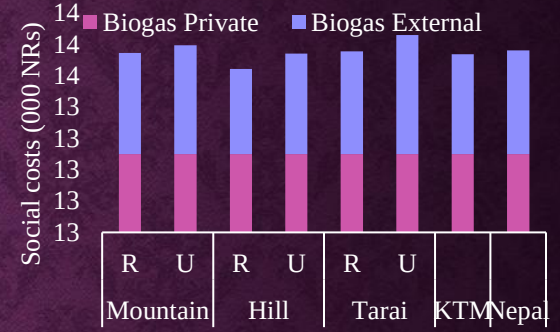
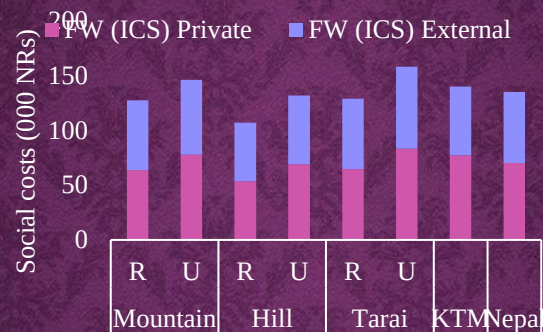
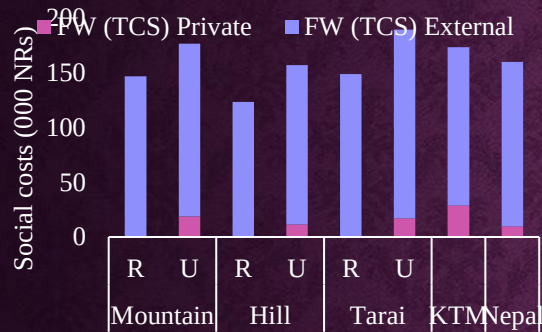
Results: Externality costs

Externality costs (000 NRs/household/year)



Results: Social costs

Social costs (000 NRs/household/year)



Wider Economic Impacts

If the LPG consumption for cooking, which is 18 PJ in 2023, is substituted with electricity (hypothetically):

It would need:

- ❖ 1,145 MW of new hydropower capacity (about one third of current capacity)
- ❖ US\$2,863.5 million investment for the construction of hydropower plants
- ❖ It would create:
 - ❖ 14,600 jobs each year during the construction period (5 years)
 - ❖ Another 2,000 jobs each year for operation of power plants (40 years)
- ❖ It would add directly annual value added of 0.4% of Nepal's GDP in 2023 in the construction sector (5 years); annual value added of 0.31% of 2023 GDP in the power sector (40 years)
- ❖ Indirect economic outputs would be 4.5 times through the construction sector and 3.5 times through the electricity sector

Conclusions

- ❖ Nepal depends on traditional biomass and imported LPG for cooking and faces economic and environmental consequences despite the potential of substituting these fuels with domestically produced clean hydropower;
- ❖ From private or user's perspective, electric cooking is found cheaper than fossil-fuel based cooking but expensive than biomass cooking, but from social perspective, electric cooking is found the cheapest among all options considered except biogas;
- ❖ Substituting imported LPG with domestic electricity is not only economic on cooking cost basis, but it also generates wider economic benefits (through the addition of new hydropower plants);
- ❖ If the LPG used in 2023 is replaced with electricity, it would require to add new electricity capacity which one-third of current capacity; the yearly value added created would be 0.4% (construction sector) and 0.3% (electricity sector) of Nepal's GDP in 2023.

THANK YOU

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